

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, SQL, JavaScript, TypeScript, C++, Java, Bash

ML & Deep Learning

PyTorch, TensorFlow, JAX, scikit-learn, NumPy, pandas, classification, regression, model selection, hyperparameter tuning, transfer learning

NLP & LLM

NLP, LLM agents, RAG, Hugging Face Transformers, sentence-transformers, prompt design, OpenAI / Anthropic / Groq APIs, evaluation

Data & Pipelines

Apache Spark / PySpark, ETL, data cleaning, feature engineering, dataset curation, PostgreSQL, MongoDB, SQL

Cloud & Tools

AWS, GCP, Azure, Docker, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to **collect, label, and curate datasets** for production clinical models; cut manual work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered a multi-tenant **NHS clinical backend** (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly patient submissions**; owned reliability and remediated **20+ VAPT findings** protecting sensitive patient data.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed **20+ REST APIs** (Node.js, Fastify, PostgreSQL) and integrated **LLM-powered** context-aware replies into a chat product, an interactive AI application, iterating on prompt design and response-quality heuristics.
- Shipped an **open-source LLM feature** to production, **4-bit quantized Llama-7B via llama.cpp**, with per-persona prompts, async queued generation, rate limits, and a content-safety filter, built end-to-end with no ML mentor.

Projects

Deep Learning & Generative Models - From-Scratch ML

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning

- Trained and benchmarked **5 neural architectures** (perceptron, MLPs, VGG-style CNN, ResNet) on **CIFAR-10** across **3 optimizers** with **learning-rate tuning** and model selection, documenting train/test-error curves.
- Built a **reverse-mode autodiff engine** from scratch in NumPy validated against JAX, derived linear-regression asymptotics and GD/SGD convergence, and implemented **normalizing-flow** and **diffusion** generative models in JAX.

Spark Data Pipeline Optimization - Big-Data Cleaning & ETL

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL | ~1.4B-row dataset

- Built and optimized a Spark **data pipeline** over a **~1.4B-row** dataset with schema-aware reads, a 6-rule **data-cleaning filter** (retaining 95.26% of records), column pruning, and partitioning, cutting end-to-end runtime **~25%**; profiled stage-level metrics in Spark UI.

AgenticSearch - Interactive AI Application (LLM Agent)

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, OpenAI / Groq, SQLite

- Built an **interactive AI application**, a multi-stage **LLM agent** that plans retrieval, extracts entities with deterministic parsers before LLM fallback, verifies outputs, and returns cell-level provenance behind a reviewer-facing UI; deployed as a live demo on **Google Cloud Run**.
- Hardened the service with **150+ automated tests**, an **evaluation harness**, and provider routing and fallback across LLM backends.

KG2RAG-Enhanced - Multi-Hop Retrieval-Augmented QA (NLP)

github.com/Deva-1903/KG2RAG-685-NLP

Python, Ollama + LLaMA-3, sentence-transformers, llama-index | UMass CS 685 Advanced NLP

- Built the **multi-view retrieval** component of a team KG-guided **RAG** pipeline for multi-hop QA, query decomposition, **dense retrieval**, and fusion via RRF, cross-encoder reranking, and MMR, lifting supporting-fact recall **+2.68%** (54.53 to 57.21) over the KG²RAG baseline on a **4,905-question** evaluation.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.